

VILA REAL AP, Portugal

WMO: 085670

Lat: 41.2742N

Lon: 7.7172W

Elev: 1843

StdP: 13.74

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.0	32.0	19.9	16.0	37.7	23.3	18.8	39.1	18.7	46.7	15.9	47.9	2.5	30	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.4	92.3	67.2	89.1	66.3	86.0	65.5	69.8	87.2	68.3	84.9	66.9	82.6	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	95.0	76.2	62.4	90.4	74.1	61.2	86.3	72.7	34.8	87.2	33.6	85.2	32.4	83.0	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.0	11.9	10.1	DB	26.4	97.6	2.1	2.3	24.9	99.2	23.7	100.5	22.5	101.8	21.0	103.4
			WB	24.4	72.7	2.1	1.6	22.9	73.9	21.7	74.8	20.5	75.7	19.0	76.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	43.7	46.0	50.4	53.4	59.4	66.3	70.4	71.2	66.8	58.7	48.8	44.2		
	DBStd	11.53	5.10	4.96	5.74	6.82	6.96	7.14	6.56	6.28	5.97	5.94	5.33	5.00		
	HDD50	717	201	128	67	38	5	0	0	0	0	5	84	189		
	HDD65	3696	659	533	452	354	202	71	22	13	49	211	486	644		
	CDD50	3153	6	16	80	139	297	488	632	658	505	275	48	9		
	CDD65	656	0	0	0	5	29	109	189	206	103	15	0	0		
	CDH74	6641	0	0	2	57	317	1147	2012	2115	885	106	0	0		
	CDH80	2539	0	0	0	3	73	416	847	902	278	20	0	0		
(14) Wind	WSAvg	3.9	3.6	4.1	4.3	4.3	4.2	4.3	4.4	4.1	3.4	3.2	3.5	3.2		
(15) Precipitation	PrecAvg	43.8	6.3	6.7	3.8	3.5	2.8	2.1	0.6	0.6	1.9	4.3	4.9	6.3		
	PrecMax	71.3	17.5	17.7	14.1	10.4	7.5	9.4	2.9	2.7	6.4	10.8	18.1	24.4		
	PrecMin	24.2	0.0	0.9	0.0	0.2	0.4	0.1	0.0	0.0	0.1	0.0	0.0	0.4		
	PrecStd	11.6	4.4	5.0	3.5	2.6	1.8	2.1	0.8	0.7	1.6	3.1	3.9	5.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	64.3	72.5	79.3	87.1	92.9	95.6	96.7	91.8	83.0	65.5	57.8		
		MCWB	51.6	51.9	54.3	60.0	64.9	68.6	67.8	68.6	65.4	60.6	56.6	53.3		
	2%	DB	54.9	60.1	68.7	75.1	81.3	88.4	92.1	92.4	86.7	76.4	61.5	55.5		
		MCWB	51.0	51.0	53.2	58.4	62.9	66.5	67.5	66.7	64.3	60.4	55.2	52.3		
	5%	DB	53.1	56.9	64.9	70.5	77.4	85.0	88.5	88.9	82.9	71.9	58.9	53.6		
		MCWB	50.2	49.2	52.2	56.5	61.1	65.7	66.9	66.1	63.1	59.1	54.0	51.1		
	10%	DB	51.4	53.9	60.6	65.6	73.3	81.0	84.7	85.2	79.4	68.2	56.7	51.7		
		MCWB	48.8	48.1	50.4	54.4	59.7	64.2	66.0	65.1	62.2	58.2	52.9	49.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.4	54.9	57.3	62.2	68.1	71.1	71.4	71.8	68.1	65.2	60.0	55.1		
		MCDB	55.7	59.7	67.7	76.3	81.6	88.1	88.9	90.8	85.5	72.9	63.1	56.4		
	2%	WB	52.4	52.6	55.1	59.4	64.9	68.9	69.7	69.3	66.2	63.0	57.3	53.3		
		MCDB	53.8	57.3	63.4	72.1	78.0	85.0	87.3	86.9	82.0	70.7	59.5	54.7		
	5%	WB	50.9	50.8	53.6	57.3	62.4	67.1	68.2	67.6	64.7	61.3	55.4	51.5		
		MCDB	52.5	55.3	61.2	68.5	74.5	82.1	84.9	84.3	79.2	68.3	57.7	53.3		
	10%	WB	49.1	49.1	51.9	55.1	60.2	65.3	66.6	66.2	63.2	59.7	53.3	49.6		
		MCDB	51.0	53.2	58.7	64.1	71.1	78.9	82.5	82.1	76.1	65.7	55.8	51.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.6	16.5	17.2	19.9	22.2	24.2	24.4	21.3	16.0	12.1	11.0		
		MCDBR	12.8	18.9	23.1	24.6	25.8	26.9	27.9	27.7	25.9	21.9	14.5	11.5		
		MCWBR	9.3	11.7	12.8	12.4	12.9	12.8	12.0	11.5	11.0	10.8	9.5	8.8		
	5% WB	MCDBR	10.0	16.0	18.9	22.8	24.0	25.8	26.8	26.3	23.4	17.0	11.3	10.2		
		MCWBR	8.3	10.9	11.3	11.9	12.6	12.7	12.2	11.8	10.5	9.4	8.6	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.310	0.341	0.345	0.353	0.356	0.344	0.356	0.344	0.335	0.308	0.296		
	taud		2.546	2.504	2.431	2.423	2.422	2.428	2.445	2.403	2.460	2.510	2.547	2.550		
	Ebn at Noon		275	287	289	294	292	291	293	287	284	274	268	266		
	Edh at Noon		23	27	33	35	36	36	35	36	31	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		570	902	1245	1602	1988	2260	2359	2102	1608	1007	646	511		
	RadStd		91	135	196	201	206	143	100	108	117	131	122	76		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +1.08	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) -272	(l) +324	(m) N/A	(46)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page